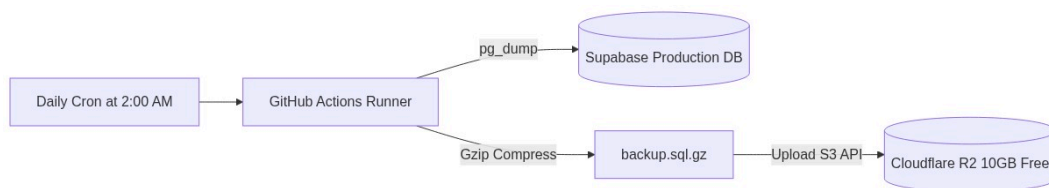


100% Free Automated Database Backup Pipeline Guide

This guide outlines how to set up a fully automated, off-site, daily database backup pipeline for your Supabase PostgreSQL database using 100% free tier services:

1. GitHub Actions (Free cron scheduler & environment runner).
2. Cloudflare R2 (10GB free cloud storage with S3-compatible API and \$0 egress/download fees).

The Workflow Architecture



Step 1: Create a Free Cloudflare R2 Bucket

Cloudflare R2 offers 10GB of storage per month completely free with zero download fees.

1. Create a free account at Cloudflare.
2. Navigate to R2 Object Storage in the sidebar.
3. Click Create Bucket, name it hiresense-db-backups, and click create.
4. In the R2 dashboard, click Manage R2 API Tokens on the right side.
5. Click Create API Token:
 - Token Name: github-backup-uploader
 - Permissions: Admin Read & Write
 - Click Create Token.
6. Copy the following credentials immediately (you won't be able to see them again):
 - Access Key ID
 - Secret Access Key
 - Jurisdiction-specific Endpoint (looks like [https://\[ACCOUNT-ID\].r2.cloudflarestorage.com](https://[ACCOUNT-ID].r2.cloudflarestorage.com))

Step 2: Configure Secrets in GitHub Repository

To protect your database credentials, save them as encrypted repository secrets in your GitHub repository.

1. Go to your repository on GitHub.

2. Click Settings Secrets and variables Actions.
3. Click New repository secret and add the following 5 credentials:

Secret Name	Value Example
DB_CONNECTION_STRING	postgresql://postgres:[REF]:[PW]@aws-0-us-east-1.pooler.supabase.com:5432/postgres
R2_ACCESS_KEY_ID	*Your Cloudflare R2 Access Key ID*
R2_SECRET_ACCESS_KEY	*Your Cloudflare R2 Secret Access Key*
R2_ENDPOINT	https://[ACCOUNT-ID].r2.cloudflarestorage.com
R2_BUCKET_NAME	hiresense-db-backups

Step 3: Create the GitHub Actions Workflow File

Create a new directory and file in your repository at `.github/workflows/db-backup.yml` with the following YAML code. This script will run automatically every day at 2:00 AM UTC and can also be triggered manually.

```
name: Daily Database Backup

on:
  schedule:
    # Runs every day at 2:00 AM UTC
    - cron: '0 2 * * *'
  workflow_dispatch:
    # Allows you to manually trigger the backup from the GitHub Actions tab

jobs:
  backup:
    runs-on: ubuntu-latest

    steps:
      # 1. Install PostgreSQL client utilities
      - name: Install pg_dump
        run: |
          sudo apt-get update
          sudo apt-get install -y postgresql-client

      # 2. Extract database backup using pg_dump
      - name: Run Database Dump
        run: |
          pg_dump "${{ secrets.DB_CONNECTION_STRING }}" -F c -b -v -f "hiresense_backup.dump"

      # 3. Configure AWS credentials pointing to Cloudflare R2
      - name: Configure AWS Credentials
        uses: aws-actions/configure-aws-credentials@v4
        with:
          aws-access-key-id: "${{ secrets.R2_ACCESS_KEY_ID }}"
          aws-secret-access-key: "${{ secrets.R2_SECRET_ACCESS_KEY }}"
```

```
aws-region: us-east-1 # Dummy region required by AWS SDK

# 4. Upload dump file to Cloudflare R2
- name: Upload to Cloudflare R2
  run: |
    aws s3 cp hiresense_backup.dump s3://${ secrets.R2_BUCKET_NAME
}}/backups/hiresense_backup_$(date +%F_%H-%M).dump --endpoint-url ${ secrets.R2_ENDPOINT }}
```

How to Restore Your Data for Free

If you ever need to restore your database, you can download the backup file for free from your Cloudflare dashboard and run:

```
pg_restore --clean --no-owner -h [NEW-SUPABASE-HOST] -U postgres -d postgres -p 5432 -v
"hiresense_backup_date.dump"
```